Cybersecurity Threats in Cloud Computing: Analysis of Threat Modeling and Risk Assessments

Rutgers University Newark, NJ

Galathara Kahanda, Ingy Omar, Yusra Elshamma

Abstract

As cloud computing becomes an increasingly integral part of individual and organizational activities, security threats to cloud infrastructure are becoming more severe. Critical security controls, such as encryption, firewalls, multi-factor authentication (MFA), and intrusion detection systems (IDS), provide minimal protection but are insufficient to address new-generation threats, including insider attacks, account hijacking, and sophisticated malware. Existing risk assessment methodologies, such as CVSS, Attack Trees, and even well-liked approaches like STRIDE, while useful, are severely limited in handling large-scale dynamic cloud infrastructures, limitations in data encryption, and weaknesses in log audits. By comparative analysis, the review contrasts threat modeling approaches for their comparison in this review across different approaches of threat modeling—STRIDE and its structured approach to threat grouping—Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege. The paper additionally addresses fuzzy logic approaches for their quality value in refining the estimation of risk through quantifiable mathematical analysis. Lastly, this review emphasizes the necessity of using flexible, adaptive threat modeling models with fuzzy logic to prioritize and counter emerging cybersecurity threats in cloud computing more efficiently.

Introduction

Cloud computing is the on-demand delivery of computing resources. Specifically, servers, data storage, networking, application development tools, and software, over the internet on a pay-as-you-go basis. It allows individuals and organizations to utilize remote servers housed in large data centers without owning or managing physical hardware. Cloud computing's flexibility and scalability have made it integral to a wide variety of applications, from everyday tasks like streaming media and email access to business processes

requiring vast amounts of computing power and storage. By leveraging cloud services, users can scale resources quickly, enable collaboration, and leverage emerging technologies without having to make significant upfront capital investments. As the adoption of cloud computing increases among organizations, security has become a major concern with respect to cybersecurity threats. But as cloud computing becomes more widely used, cybersecurity is becoming an important matter. As more businesses shift their activities to the cloud, they must contend with a constantly changing array of security risks. These risks include data outsourcing that leads to malicious attacks, human error leading to data loss, account hijacking, DDoS attacks, misconfiguration, API vulnerabilities, and malware/ransomware. All these pose serious risks to sensitive information and businesses. Although there are risk models in place, a lot of these traditional risk models struggle to face the evolving threats. Encrypted data limits functionality, Log audits are insufficient and can be tampered with, and auditors themselves can be compromised. Among these difficulties is the urgent need for strong cybersecurity measures that can protect cloud systems from evolving threats. These limitations require effective risk model assessments, such as fuzzy set analysis. This paper discusses the various security controls in place, how threat modeling helps identify these threats, and how risk assessments help identify which model is the best and which threats need the most attention. Our review concludes that Stride and fuzzy logic help improve risk prioritization the most.

Cybersecurity Threats in Cloud Computing

Research by Awodele evaluated cloud security threats and assigned risk values based on likelihood and impact. Their analysis found that data breaches and insider threats ranked among the most severe threats in cloud computing. Some of the common threats in cloud computing are data breaches, which are one of the most serious threats to cloud systems, caused by the loss of encryption keys and the disclosure of private information. They are positioned as high-risk to negatively impact organizations. Account Hijacking is also a significant risk,

especially if passwords are overused or weak. Given that impersonation and unauthorized access to login credentials are common attack routes. There are also Insider threats. In many situations, malicious insiders pose a significant risk, particularly when cloud services are being threatened, especially when proper inspection is lacking. Because insiders have privileged access within an organization, it can be challenging to identify insider threats. Lastly, Insecure Interfaces and APIs. Because attackers can take advantage of poorly designed or exposed interfaces and APIs, this poses a significant threat to cloud cybersecurity. Under factors like misplaced encryption keys, illegal access to login information, and misuse of cloud services.

Current Security Controls and Their Limitations

A survey on cloud security requirements, threats, vulnerabilities, and countermeasures by Rakesh Kumar and Rinkaj Goyal covers security measures to reduce dangers. Security tools like encryption, firewalls, access limits, multi-factor authentication (MFA), as well as intrusion detection systems (IDS), are important for protecting cloud infrastructure from possible weaknesses.

Encryption is a fundamental step for the protection of sensitive data, both when moving and when stored. It changes readable data into a coded form. It makes certain that people who should not have access cannot read the data. In cloud computing, this is especially important for guarding personal, financial, and intellectual property data stored in the cloud. Although they face performance difficulties such as “poor key management, faulty, insecure and obsolete encryption algorithm” (pg.15), Data encryption and decryption can slow down system performance. This is especially true when dealing with large datasets or cloud environments with larger data movement. As well as managing encryption keys, which can be tough. Should keys get lost or are badly managed, it poses a risk to data availability and usability.

Firewalls also act as a crucial barrier between cloud systems and external threats. They use pre-established security rules to filter network traffic coming in and going out. They aid in preventing unwanted

access, especially from threats like DDoS assaults or illegal attempts to access resources. However, because cloud environments are dynamic, firewalls have limitations. Vulnerabilities arise from the frequent resource changes in cloud deployments. “Traditional data center perimeter-security measures like firewalls... are not enough to secure the virtual machines” (p.14). Because of this, maintaining static firewall rules is challenging. This modification may cause defense flaws. Firewalls cannot protect against internal threats like authorized users abusing their access privileges.

Access control systems allow only authorized users to reach private cloud resources. As defined by the survey, it is “the security discipline that enables the right individuals to access the right resources at the right times for the right reasons.”(pg.17) Role-based access control (RBAC), attribute-based access control (ABAC), as well as other methods, are commonly used to let users reach only the data they require. Access controls can have weaknesses because of setup errors. For example, giving users privileges or roles can unintentionally grant too much access. This exposes private data to unauthorized persons. Even robust access controls do not defend against stolen credentials. If an attacker steals valid user credentials, they can avoid access controls entirely.

Moreover, MFA needs users to give two or more proof types before they get to cloud items “by using hard tokens, soft tokens, out-of-band passwords, and biometrics”(pg.17). MFA works well to lower the danger from access done without permission because of weak or stolen passwords. Social engineering attacks, such as phishing, can get around MFA. An attacker may fool a user into giving authentication details. Not only that, but MFA can make usability hard. Users may need to authenticate a number of times or handle a number of authentication methods, which many users avoid taking the extra precaution because it is time consuming.

Finally, IDS has a job in identifying possible security breaches. It watches network traffic and system behaviors for malicious activity. IDS systems find a variety of attacks. DDoS, malware, as well as unauthorized access attempts. This permits a fast reaction and mitigation. Often, they have false positives when good activities seem suspicious. False negatives also happen when actual attacks are missed. It is not used as a preventative, as IDS often reacts after an attack is seen, rather than stopping it before it happens.

Firewalls, access controls, multifactor authentication, as well as intrusion detection systems are parts of cloud environment security. Those security systems alone do not always address the changing difficulty of cloud security threats. It is challenging to offer reliable and adequate protection using traditional security measures like firewalls or access lists because of these limitations. The majority of traditional controls were created for on-premises IT systems and are not equipped to handle the complexity of the shared responsibility model of cloud computing.

Threat Modeling & Risk Assessment

Security control alone is not effective enough to battle against threats. Establishing a blueprint before executing a practice can help improve the chances of mitigating threats immensely. Which is why threat models exist. Threat modeling as described by Bodeau, McCollum, and Fox (2018) is a structured process used to uncover, examine, and mitigate cybersecurity threats at various levels, including system, business, organizational, and enterprise levels. At the system levels, it is used to guide security control decisions in operations and design. At the organizational level, it reflects the organization's assumptions about its threat environment and influences risk management decisions, including SLAs with service providers like cloud services. At more aggregated levels, threat modeling facilitates the sharing of threat intelligence and supports

collaborative efforts like cyber wargames. Overall, it constitutes a valuable component of proactive risk analysis and serves to ensure security is set in throughout the system and the organization's life cycle.

Some of the popular techniques are STRIDE, for example, according to Naik et al. (2024), STRIDE divides threats into six categories: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege. STRIDE is effective in addressing spoofing and tampering concerns. It mainly focuses on the part where data is moving in the cloud, which will help discover any vulnerabilities much more easily (pg.271). Attack trees are used to determine how an attacker would attempt to compromise a system. They make it simpler to evaluate risk or launch simulated cyberattacks (also known as cyber wargaming) by illustrating the various ways an attack could occur. PASTA (Process for Attack Simulation and Threat Analysis) is another technique that evaluates cloud security. Pasta has phases to fully cover a system. The first phase is analyzing businesses' applications, then the hardware and software they are using. The second phase is a deeper dive into the data flow and data logs in these applications and detect vulnerabilities and possible threats. Third phase, they would do an attack simulation to identify all the areas needing security control and then establish a security strategy for the business (pg.80). LINDDUN is a framework for privacy risk analysis.Gained recognition because of its capacity to identify privacy issues in cloud systems through identifying privacy threats such as Linkability, Identifiability, Nonrepudiation, Detectability, Disclosure of information, Unawareness, and Noncompliance (pg.283).

Fuzzy Logic & Quantitative Risk Evaluation

To determine which threat model is the best, Awodele (2024) used a fuzzy set method for a risk assessment. Fuzzy set theory is a mathematical framework that can handle extreme ambiguity and vagueness in data. Their rationale for doing so is based on threat modeling, which makes it possible to examine a system's architecture, determine potential threats, and develop mitigation plans. The threat modeling process was

conducted by sampling cloud infrastructures with the overall goal of helping experts develop a cyberthreat mitigation model that can be used in cloud environments. In their research applied fuzzy set theory to the analysis of cloud security risks' uncertainties and discussed various mitigation strategies. The fuzzy values range table (Awodele et al. 2024, Table 1) was created to detect different risk tiers easily, from low to very high. The authors established specific value sets (0.1-1) and (1-5). This lets them convey the challenges present in assessing cybersecurity hazards. The use of fuzzy values allows a clearer quantification of risks. This also aids organizations to better assess as well as react to these risks.

		Threat Mitigation Techniques											
S/N	Features	STRIDE	QTMM	CVSS	PASTA	PnG	Attack Tree	HTMM	VAST Modelling	LINDDUS	OCTAVE	Trike	Security Cards
1	Documentation	5	1	2	4	2	2	4	4	4	2	2	1
2	Techincal Threat identification	5	5	5	5	5	5	2	2	3	3	2	5
3	Non technical threat identification	5	5	5	3	3	5	4	4	3	4	4	3
4	General threat identification	5	5	5	5	3	5	5	3	4	3	4	3
5	Time consumption	2	3	3	1	4	3	4	1	2	4	1	3
6	Usage	5	4	4	1	5	4	3	5	3	4	3	3
7	Model maturity	5	5	3	3	3	3	4	3	4	4	3	2
8	Training/usage requirments	5	4	4	2	3	4	4	3	3	3	4	4
9	Business impact	1	3	3	5	3	1	2	4	2	4	3	1
10	Security properties	5	5	3	4	3	4	5	3	4	4	4	3
11	Threat classification	5	4	4	3	3	3	3	3	3	1	3	4
12	Stakeholders input/ collaboration	4	4	1	5	3	1	1	3	3	2	3	5
13	Threat prioritization	3	5	5	5	4	3	3	3	4	2	3	4
14	Reliability	4	4	4	4	3	3	2	4	3	4	4	1
15	Total	59	57	51	50	47	46	46	45	45	44	43	42

There were several threat modeling methods analyzed, including STRIDE, PASTA, LINDDUN, CVSS, Attack Tree, PnG, Security Cards, HTMM, QTMM, Trike, VAST Modeling, and OCTAVE. To support their analysis, the authors performed a comparative analysis of such methods of mitigation and assessed them according to 14 key attributes, ranging from the quality of documentation, technical ability in detecting threats, efficiency in time, to usefulness in diverse cloud scenarios. Showing that the STRIDE methodology is the optimal strategy.

Not only is this method able to analyze risk assessments, but it is also able to analyze which cybersecurity threat poses the most risk. The table outlines the most common cybersecurity risks to cloud

environments and simplifies them to data breaches, insider threats, account hijacking, DDoS attacks, misconfiguration, API vulnerabilities, and malware/ransomware as shown in table 3 (Awodele et al. 2024, Table 3) is a quantitative assessment of risk for various scenarios and then generates an overall risk score for each danger. For instance, the threat of data loss from natural disasters has been given a fuzzy value of 0.75, which indicates a relatively high probability, while the value of 0.25, which indicates a lesser risk in this context, has been allocated to account hijacking by impersonation. In order to rank security measures, there is a total at the end of the table of the risk severity across all categories. This table is a key tool for converting qualitative risk data into taking action and ensuring that mitigation strategies match what needs to be prioritized.

Table 3: Analysis of cybersecurity threats using fuzzy set analysis

		Cybersecurity Threat Features										
S/N	Cause	Data Breaches	Account Hijacking	Data Loss	Malicious Insiders	Abusive Use of Cloud Services	Insecure Interfaces and APIs	Insufficient Due Dilligence	Identity Theft	Shared Technology Issues	Changes in Business Model	Lock-in
1	Natural Disaster	0.25	0.25	0.75	0.25	0.25	0.25	0.25	0.25	0.25	0.75	0.25
2	Loss of encryption Key	0.75	0.5	0.75	0.25	0.25	0.75	0.25	0.25	0.25	0.25	0.25
3	Illegal access to login details	0.75	0.75	0.25	0.25	0.25	0.25	0.25	0.75	0.25	0.25	0.25
4	Undeserved operations conducted by users	0.75	0.75	0.75	0.75	0.25	0.25	0.25	0.25	0.25	0.25	0.25
5	Unknown cloud storage	0.75	0.25	0.25	0.75	0.25	0.25	0.75	0.25	0.25	0.25	0.75
6	Exposure to sensitive information	0.75	0.25	0.25	0.75	0.25	0.25	0.25	0.25	0.25	0.25	0.25
7	Unlawful acts of users	0.75	0.25	0.75	0.75	0.75	0.25	0.25	0.25	0.25	0.25	0.25
8	Usage of the same VM by multiple users	0.25	0.75	0.25	0.25	0.75	0.25	0.25	0.75	0.75	0.25	0.25
9	Impersonation	0.25	0.75	0.25	0.25	0.25	0.25	0.25	0.75	0.75	0.25	0.25
10	Operating system failure	0.75	0.75	0.75	0.25	0.25	0.25	0.25	0.25	0.25	0.75	0.25
11	Following internal security measures	0.75	0.75	0.75	0.25	0.25	0.25	0.75	0.25	0.25	0.25	0.25
12	Total	13.5	12	11.5	9.5	7.5	6.5	4.5	4.5	4	4	3.5

Documentation through AVAST and Angurag visual shows that in 2012, the number of executable malware by attackers was 100 million, and in 2022, this number has reached an astronomical height of 1.9 billion. (Awodele et al. 2024).These numbers are pushing towards the conclusion that attackers are willing to invest significant time and effort into cloud cyber attacks because they believe that it's feasible and executable. Considering that in 2022, data breaches alone cost companies 8.19 million USD worldwide (Awodele et al.

2024), these statistics prove that cloud security is still lacking and unable to keep up with the surge of attacks that are constantly being produced. Furthermore, the amount of capital to be gained from successfully carrying out an attack against cloud infrastructures is enough to motivate attackers to stay persistent with their attempts, as they stand to gain a lot if they are successful. These points have prompted researchers into divesting a lot of energy into researching ways to improve cloud security as it stands and one of the categories they have been exploring on how to improve is threat modeling frameworks.

Cyber threats to cloud computing are adapting and becoming too sophisticated for traditional threat modeling methods to handle. Even though models like STRIDE, PASTA, LINDDUN, and OCTAVE have all been very effective in deterring cyber attacks so far, they all share one limitation in common: they're all static (Kharma and Taweel). Being static means that, based on predefined knowledge and attacks that have already been seen or recorded, this contradicts the dynamic nature of cloud environments, which are constantly changing and adapting, including shifts in data centers, intrusion detection systems, third-party vendors, APIs, and numerous other variables.

New threats against cloud computing arise as time progresses because cloud computing is constantly evolving, and with this evolution, there should be evolving threat modeling methods to account for all the changes that are occurring.

Table 7

CSA top threats to cloud computing			
Top Threats	2010	2013	2016
1	Abuse and nefarious use of cloud computing	Data breaches	Data breaches
2	Insecure application programming interfaces	Data loss	Weak identity, credential and Access management
3	Malicious insiders	Account hijacking	Insecure APIs
4	Shared technology vulnerabilities	Insecure APIs	System and Application Vulnerabilities
5	Data loss/Leakage	Denial of service	Account hijacking
6	Account, Service and Traffic hijacking	Malicious Insiders	Malicious Insiders
7	Unknown Risk Profile	Abuse of Cloud Services	Advanced Persistent Threats (APTs)
8	----------	Insufficient Due Diligence	Data loss
9	----------	Shared Technology Issues	Insufficient Due Diligence
10	----------	----------	Abuse and Nefarious Use of Cloud Services
11	----------	----------	Denial of service
12	----------	----------	Shared technology vulnerabilities

This figure, presented by Kumar and Goyal, shows the top cloud computing threats over the years by the CSA, Cloud Security Alliance, an organization that aims to homogenize cloud security standards, creating healthy security practices that ensure cloud confidentiality. CSA obtained this information by surveying a variety of individuals who work in the field of cloud confidentiality As seen in the figure the list of top 10 threats keeps changing over the years, some threats got completely eliminated like unknown risk profiles, and some new threats were introduced like denial of service and insufficient due diligence. This goes to show the dynamic nature of the cloud and how attack vectors and surfaces are constantly being produced against its environments, and this also proves that cloud security is evolving, which is proven by the disappearance of

some of the threats seen in 2010. The evolution of threat prioritization shown in the table calls for a proactive approach to threat modeling that is able to adapt to these ever-changing variables, incorporating things like live threat intelligence, automation, and flexible modeling components.

As of today, STRIDE remains one of the most widely used threat modeling techniques in the field of cybersecurity, This could be attributed to its ease of use compared to other threat modeling techniques. Naike et al. provide a comparative overview of six of the most common threat modeling methodologies, STRIDE, DREAD, VAST, PASTA, OCTAVE, and LINDDUN, across multiple features, including scalability, applications, and ease of use. STRIDE is the only method in the study rated as 'easy' to use, supporting the argument that its simplicity is a key factor in its widespread adoption and implementation, particularly among small and fast-paced development teams. This is also further supported by the fuzzy set analysis done by Adwole et al, which showed that compared to the other threat modeling methodologies, STRIDE scored higher than all the other methodologies, showcasing its many benefits that make it appealing to developers. Despite STRIDE's many redeeming features, it is still often insufficient on its own; this could be attributed to its loosely structured processes of application. This causes concerns in environments where STRIDE is being utilized by inexperienced developers who prioritize speed and meeting deadlines over security concerns. This would open the way for them to apply STRIDE more as a checklist of requirements rather than a threat model, which would cause many critical and dangerous threats to go overlooked.

Case Studies: Evaluating STRIDE in Practice

Two case studies provide empirical insight into how STRIDE is applied and evaluated in academic settings. The first, by Scandariato et al. (2015), and the second by Landuyt and Joosen (2021). Both studies involved graduate students and security experts as participants. Each provided participants with instructions on

STRIDE through lectures and reading materials to ensure a shared baseline of understanding. The 2015 study aimed to assess the effectiveness of STRIDE by means of quantifiable data such as the number of true positives (valid threats), false negatives (threats missed), and false positives (invalid threats) produced by the students. The 2021 study, on the other hand, aimed at uncovering the number of assumptions made by participants and the effect these assumptions had on the threats that were generated. The 2015 study was part of an elective graduate course, conducted in a collaborative environment with students working in teams of three or four. The work of this study was done across their semester, and the resulting threats were evaluated by two experts, who provided feedback to improve the students' understanding and application of STRIDE. The 2021 study was conducted in an examination setting and was done individually by students and 5 experts alike.

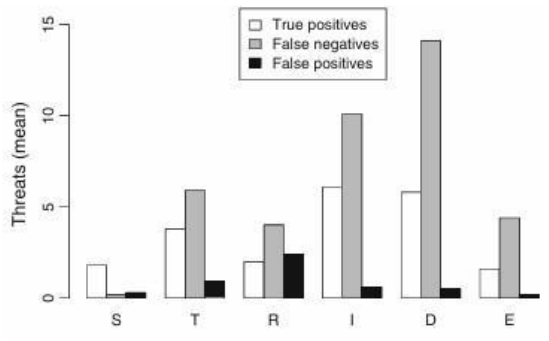

Fig. 8 True positives, false negatives and false positives by threat category

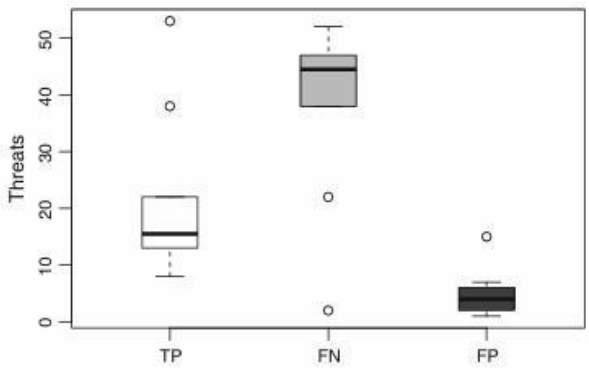

Fig. 7 *Boxplot* of true positives, false negatives and false positives

In the 2015 study, the results were as seen in Figures 8 and 7. Per their analysis, the average number of false negatives was 64-69%, and the denial of service category had the highest number of overlooked threats among all the other categories of STRIDE. A contributing factor as to why there is a significant number of false negatives is the fact that this study was conducted via labs on pen and paper and didn't utilize any tools or automate any tasks, making these results completely dependent on the student's knowledge and interpretation of the applications of STRIDE. This raises an important question: Do these false negatives indicate a limitation of

the STRIDE model itself, or do they reflect insufficient training on STRIDE from the students? Considering that Scandariato et al described the amount of training their students had as "higher than the standard at Microsoft, where it is reported that "most practitioners have at most 2 h of threat modeling training"(Scandariato et al,2015), this would refute the assumption that it was the quantity of training and but rather how effectively students applied STRIDE during the exercises. These findings illustrate the importance of evaluating not only the methodology but also the inherent limitations of STRIDE when applied in low-tech, whiteboard settings.

The 2021 study aimed to assess the effect assumptions had on the threats being produced. Through their study, they assigned a T_ROLE variable that participants use for their assumptions to document the role that assumption plays in their threat modeling process. As seen from the table, the majority of assumptions by students and experts (58% of students, 73% of experts) aimed to justify the exclusion of threats, and the

Table 4: The results of applying T_ROLE

T_ROLE	Student assumptions	Expert assumptions
Exculde	351 (58.31%)	28 (73.68%)
Include	144 (23.92%)	7 (18.42%)
Prioritize	95 (15.78%)	1 (2.63%)
Undetermined	12 (1.99%)	2 (5.26%)

Documentation for their reasoning behind these assumptions was deficient, as described by Landuyt and Joosen. These findings highlight the importance of proper documentation for threat exclusion justification, as this could be detrimental as products and software progress through their development cycles. Furthermore, a possible explanation behind the inadequate documentation for assumptions meant for threat exclusion could lie in the fact that when using STRIDE, the participants had to progress through the data flow diagram nodes individually, and threat exclusion is done early on to avoid threat explosion when listing possible threats to DFD nodes. Another factor that could be contributing to these findings is that this study was conducted in an exam

format where the students and the experts were given 2.5 hours to complete the exercise on pen and paper and were not allowed access to the internet or outside sources. The reasoning behind that format is that Landuyt and Joosen feared outside influence would affect their study negatively. This fast paced environment replicates real conditions faced by developers in the industry and illustrates the risks of accelerating the threat modeling phase, and showcases that time pressure can lead to the exclusion of valid threats in favor of rapid delivery.

Abuabed et al conducted their own study in 2023, where they combined the use of STRIDE for threat modeling, CVSS for evaluating exploitability, and attack trees for modeling attacker strategies to do threat prediction and risk assessment on an advanced driver assistance system (ADAS). ADAS controls the navigation system, lane keeping system, adaptive cruise control, and anti lock braking system for a smart car model.

Through their study, Abuabed et al utilized tools such as MS TMT and expanded on it with their own frameworks that they coded using Java to help them automate the threat prediction process, along with automating the analytical scoring for risk matrices, impact scoring, and feasibility scoring. Abuabed et al extended the MS TMT framework so that it could support non-expert use and provide automated guidance to new developers. MS TMT generates threats automatically based on the structure of a data flow diagram (DFD), using STRIDE threat categories associated with each DFD node. Furthermore, it provides a field for documenting assumptions that do not influence threat generation directly, but could be crucial in the assumption documentation process. As with MS TMT, the accuracy of the results is heavily dependent on the expertise of the person operating it. If key architectural elements are misrepresented or assumptions are incorrect or missing, the generated threat model may be incomplete or misleading, causing many threats to go unnoticed or causing the developers to put time and energy into defending against threats that are inconsequential. This highlights a recurring theme seen across the literature, and that is the importance of using threat modeling methodologies by

experienced individuals. These individuals need to have a background in cybersecurity principles as well as have a thorough understanding of the system under development; the absence of one of these qualities could lead to oversights in system security. Although Abuabed et al extended the MS TMT framework, their work underscored the broader issue, which is the unavailability of beginner-friendly tools and methodologies that account for scenarios of improper use. While STRIDE and MS TMT often promote themselves as being easy to learn and beginner-friendly, they don't discuss the misuse case scenarios they face and the high likelihood of their improper use if handled by an inexperienced individual. Another important point worth noting is that Microsoft has never published any data that discusses the effectiveness of STRIDE, even though the technique is often praised for its ability to improve product quality (Scandariato et al., 2015)

Theme Across Literature

Throughout the literature, there has been a consistent criticism of STRIDE by several authors who highlight its ad hoc nature and susceptibility to improper application. In this context, “ad hoc” refers to threat modeling practices that are performed in an unstructured or situational manner, often lacking formal process or repeatability, making it vulnerable to variability in application depending on the user’s experience. Yskout et al, as cited by Kharma and Taweel (2023), suggested a five-level maturity model for threat modeling frameworks: Level 1 (Ad hoc), Level 2 (Repeatable but intuitive), Level 3 (Defined process), Level 4 (Managed and measurable), and Level 5 (Optimized). Although STRIDE is widely regarded as easy to learn and practically effective, it has remained at ad hoc maturity since its introduction in 1999. This lack of evolution since its introduction makes its applicability limited in cloud computing and dynamic environments. As a result, STRIDE is vulnerable to emerging threat vectors and complex architectural models of development, ultimately reducing its effectiveness as a threat modeling framework in today's rapidly evolving technology landscape.

One of the major contributors to STRIDE's stagnation, noted by multiple authors, is its dependence on data flow diagrams (DFDs). DFDs have been described by the authors as being time-consuming, agnostic to security measures, and limited in their ability to express the complexity of system behaviors (Abuabed et al, 2023; Landuyt & Joosen, 2021; Scandariato et al, 2015). These limitations, along with its loose documentation practices, contribute to STRIDE's continued classification as an ad hoc methodology. One solution offered by Pandi et al was their proposal of combining the use of STRIDE along with the pervasive computing paradigm: an approach that reflects how modern systems like smart vehicles and IoT devices are constantly connected, data-driven, and embedded into real-world environments (Kharma and Taweel,2023). This concept is reflected in the study conducted by Abuabed et al, who incorporated CVSS, STRIDE, and attack trees along with using automated tools such as MS TMT and custom frameworks that they developed for a more automated structure of threat analysis. Though the results aren't as quantifiable as in the studies conducted by Scandariato et al. or Landuyt and Joosen, their automation process aided in the streamlining of the threat prediction stage and a reduction in the over-prioritization of inconsequential threats. This addresses one of the challenges faced by Scandariato et al., Landuyt, and Joosen, where threat prediction was conducted manually through whiteboard-based sessions. By systematically reviewing, automating, and refining the threats generated through MS TMT and their custom toolset, Abuabed et al. demonstrated a more structured and adaptive approach that aligns with modern computing needs and that is more beginner and novice-friendly.

Conclusion

While further analysis is needed to completely evaluate the effectiveness of STRIDE in cloud computing environments, the literature is illustrating many deficiencies that could become exploitable attack surfaces by

malicious agents. The limitations of STRIDE include it lose documentation practices, static nature, dependance of data flow diagrams, loose structure that introduces variability of application and vulnerability to misapplication by both inexperienced users and experienced one in fast paced environments.Until research of STRIDE application progresses it is best to exercise caution with its use and supplement it by utilizing automated tools, as well as combining STRIDE with other threat modeling frameworks. This more layered and automated application of STRIDE can help address the dynamic and evolving threat landscape of cloud environments, by that means strengthening the protection of sensitive user and governmental data.

References:

Awodele, O., Olayemi, A. B., Fashoto, S. G., Balogun, F. O., & Adewumi, A. O. (2024). *Characterization and risk assessment of cybersecurity threats in cloud computing: A comparative evaluation of mitigation techniques.* Acadlore Transactions on AI and Machine Learning, 3(2). https://www.acadlore.com/article/ATAIML/2024_3_2/ataiml030204

Bodeau, D. J., McCollum, C. D., & Fox, D. B. (2018, April 7). *Cyber threat modeling: Survey, assessment, and representative framework*. Prepared for the Department of Homeland Security, The Homeland Security Systems Engineering and Development Institute (HSSEDI), Operated by The MITRE Corporation.

Naik, N., et al. "A Comparative Analysis of Threat Modelling Methods: STRIDE, DREAD, VAST, PASTA, OCTAVE, and LINDDUN." *Contributions Presented at The International Conference on Computing, Communication, Cybersecurity and AI, July 3–4, 2024, London, UK*, edited by N. Naik, P. Jenkins, S. Prajapat, and P. Grace, C3AI 2024, vol. 884, *Lecture Notes in Networks and Systems*, Springer, Cham, 2024, pp. [page numbers]. https://doi.org/10.1007/978-3-031-74443-3_16.

Kumar, Rakesh, and Rinkaj Goyal. *On Cloud Security Requirements, Threats, Vulnerabilities and Countermeasures: A Survey*. University School of Information, Communication and Technology (USIC&T), Guru Gobind Singh Indraprastha University, 2019. ResearchGate, https://www.researchgate.net/publication/333260803.

Qian, Ling, et al. "Cloud Computing: An Overview." *Cloud Computing*, edited by Martin Gilje Jaatun, Gansen Zhao, and Chunming Rong, Springer, 2009, pp. 626–631. Lecture Notes in Computer Science, vol. 5931. https://doi.org/10.1007/978-3-642-10665-1_63.Harvest

Zhang, Qi, Lu Cheng, and Raouf Boutaba. "Cloud Computing: State-of-the-Art and Research Challenges." *2010 IEEE International Conference on Cloud Computing*, IEEE, 2010, pp. 1–10. IEEE Xplore, https://doi.org/10.1109/COMSWA.2010.5583510.

Van Landuyt, Dimitri, and Wouter Joosen. “A Descriptive Study of Assumptions in STRIDE Security Threat Modeling.” *Software and Systems Modeling*, vol. 21, no. 6, 2022, pp. 2311–28, https://doi.org/10.1007/s10270-021-00941-7.

Scandariato, Riccardo, et al. “A Descriptive Study of Microsoft’s Threat Modeling Technique.” *Requirements Engineering*, vol. 20, no. 2, 2015, pp. 163–80,

https://doi.org/10.1007/s00766-013-0195-2.

Kharma, Mohammed, et al. “Threat Modeling in Cloud Computing - A Literature Review.” *Ubiquitous Security*, vol. 1768, Springer, 2023, pp. 279–91,

https://doi.org/10.1007/978-981-99-0272-9_19.

Abuabed, Zaina, et al. “STRIDE Threat Model-Based Framework for Assessing the Vulnerabilities of Modern Vehicles.” *Computers & Security*, vol. 133, 2023, pp. 103391-, https://doi.org/10.1016/j.cose.2023.103391.

Xiong, Wenjun, and Robert Lagerström. “Threat Modeling – A Systematic Literature Review.” Computers & Security, vol. 84, 2019, pp. 53–69,

https://doi.org/10.1016/j.cose.2019.03.010.

Yskout, Koen, et al. “Threat Modeling: From Infancy to Maturity.” *2020 IEEE/ACM 42nd International Conference on Software Engineering: New Ideas and Emerging Results (ICSE-NIER)*, ACM, 2020, pp. 9–12, https://doi.org/10.1145/3377816.3381741.

Khalil, Shaymaa Mamdouh, et al. “Threat Modeling of Cyber-Physical Systems - A Case Study of a Microgrid System.” *Computers & Security*, vol. 124, 2023, pp. 102950-,

https://doi.org/10.1016/j.cose.2022.102950.

Kim, Kyoung Ho, et al. “STRIDE‐based Threat Modeling and DREAD Evaluation for the Distributed Control System in the Oil Refinery.” *ETRI Journal*, vol. 44, no. 6, 2022, pp. 991–1003, https://doi.org/10.4218/etrij.2021-0181.

Abbas, Haider Hadi, et al. “Survey on Enhanced Security Control Measures in Cloud Computing Systems.” *Journal of Physics. Conference Series*, vol. 1878, no. 1, 2021, pp. 12004-, https://doi.org/10.1088/1742-6596/1878/1/012004.

Emmanni, Phani Sekhar. "Federated Learning for Cybersecurity in Edge and Cloud Computing." *International Journal of Computing and Engineering*, vol. 2, no. 1, 2021, pp. 14–25. https://doi.org/10.47941/ijce.1829.

Straub, Jeremy. "Modeling Attack, Defense and Threat Trees and the Cyber Kill Chain, ATT&CK and STRIDE Frameworks as Blackboard Architecture Networks." *Proceedings of the 2020 IEEE International*

Conference on Smart Cloud (SmartCloud), 2020, pp. 148–153. IEEE, https://doi.org/10.1109/SmartCloud49737.2020.00035.

Igwenagu, Udochukwu ThankGod Ikechukwu, and Salami, A. A. "Securing the Digital Frontier: Strategies for Cloud Computing Security, Database Protection, and Comprehensive Penetration Testing." *Journal of Engineering Research and Reports*, vol. 22, no. 3, 2023, pp. 1–15. https://doi.org/10.9734/jerr/2023/v22i31762.

Urias, Vincent E., Brian P. Van Leeuwen, William M. S. Stout, and Han Lin. "Applying a Threat Model to Cloud Computing." *Proceedings of the International Carnahan Conference on Security Technology (ICCST)*, 22–25 Oct. 2018, Montreal, QC, Canada, pp. 1–5. IEEE, https://doi.org/10.1109/CCST.2018.8585471.

Jackson, David, Barbara Harris, and Richard Clark. "Enhancing Cybersecurity Posture: A Framework for Anomaly Detection in Cloud Computing Environments." *International Journal of Information Engineering and Science*, vol. 1, no. 3, Aug. 2024, pp. 1–15. https://doi.org/10.62951/ijies.v1i3.66